

1. A Experiment with a Deeper Model and Corresponding Regularization Adjustment:

A. it involves adding an extra neural layer and increasing the number of perceptrons in previous layers.

although increasing the neural network's complexity initially reintroduces overfitting, this can be successfully mitigated by adjusting the regularization.

B. by further increasing the dropout rate, the model successfully achieves an excellent balance between the training error and the test error, ultimately resulting in a deep model with strong generalized capacity.

2. The Black Box Nature of Deep Learning:

we can train neural networks that deliver highly impressive results, but we cannot easily understand the specific mechanisms of their internal operations.

consequently, the primary focus has shifted away from understanding "how it works" to evaluating "how well it works".